7. Diabetes Technology: Standards of Care in Diabetes—2026

American Diabetes Association Professional Practice Committee for Diabetes*

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The American Diabetes Association (ADA) "Standards of Care in Diabetes" includes the ADA's current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA's clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

Diabetes technology is the term used to describe the hardware and software that people with diabetes use to assist with self-management, ranging from lifestyle modifications to glucose monitoring and automated therapy adjustments. Historically, diabetes technology has been divided into two main categories: insulin administration (insulin given by syringe, pen, inhalation, patch devices, or pump [also called continuous subcutaneous insulin infusion]) and glucose measurement (assessment by blood glucose monitoring [BGM] or continuous glucose monitoring [CGM]). Diabetes technology now also includes automated insulin delivery (AID) systems that use CGM-informed algorithms to modulate insulin delivery. Furthermore, diabetes technology encompasses connected insulin pens and diabetes self-management support software that serve as medical devices. Diabetes technology, coupled with education, follow-up, pharmacotherapy as needed, and support, can improve the lives and health of people with diabetes; however, the complexity and rapid evolution of the diabetes technology landscape can also be a barrier to implementation for people with diabetes, their care partners, and the health care team.

GENERAL DEVICE PRINCIPLES

Recommendations

7.1 Diabetes devices should be offered to people with diabetes. **A**

7.2 The type(s) and selection of devices should be individualized based on a person's specific needs, circumstances, preferences, and skill level. In the setting of an individual whose diabetes is partially or wholly managed by someone else (e.g., a young child or a person with cognitive impairment or dexterity, psychosocial issues, and/or physical limitations), the caregiver's skills and preferences are integral to the decision-making process. **E**

7.3a When prescribing a continuous glucose monitoring (CGM) device, ensure that people with diabetes and caregivers are offered initial and ongoing training and education as indicated by individual circumstances. Education should

**A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at https://doi.org/10.2337/dc26-SINT.*

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include utilization of data, including uploading or sharing data to monitor and adjust therapy. **C**

7.3b When prescribing an automated insulin delivery (AID) system, people with diabetes and their caregivers must receive education on how to use and troubleshoot the system. This education should occur at regular intervals as needed. Education should include utilization of the integrated system and its data, including uploading or sharing data to monitor and adjust therapy. **C**

7.4 Health care professionals working with people with diabetes should be aware of available technologies and seek additional support when needed. **E**

7.5 People with diabetes using CGM, continuous subcutaneous insulin infusion (CSII), and/or AID for diabetes management should have continued access to devices across third-party payors, regardless of age or A1C levels. **E**

7.6 Children and adolescents should be supported at school in the use of diabetes technology, such as CGM systems, CSII, connected insulin pens, and AID systems. **E**

7.7 For adults with diabetes using diabetes technology, reasonable accommodations in educational and work settings should include having sufficient time to manage their devices and respond to high and low glucose levels. **E**

7.8 Consider early initiation, including at diagnosis, of CGM, CSII, and AID depending on a person's or caregiver's needs and preferences. **C**

7.8a There should be no requirement of C-peptide level, **B** the presence of islet autoantibodies, **B** or duration of insulin treatment **C** before initiation of CSII or AID.

7.9 Standardized reports for all CGM, CSII, AID, and connected insulin devices with a minimum of a single-page report, such as the standardized CGM report and weekly summary, should be available and utilized. Options for daily and weekly reports and raw data should be available. **E**

Technology is rapidly changing, and there is no one-size-fits-all approach to technology use in people with diabetes. Insurance coverage can lag behind device availability, people's interest in devices and willingness for adoption can vary, and health care teams may have challenges in keeping up with newly released technology. An American Diabetes Association resource, which can be accessed at diabetes.org/living-with-diabetes/treatment-care/diabetes-technology-guide, can help health care professionals and people with diabetes make decisions on the initial choice of device(s). Other sources, including device manufacturers, can help people troubleshoot when difficulties arise (1–7).

Education and Training

In general, no device used in diabetes management works optimally without education, training, and ongoing support. There are multiple resources, including online tutorials and training videos as well as written material, on the use of devices. People with diabetes and their caregivers vary in comfort level with technology, and some prefer in-person training and support. Those with more education regarding device use have better outcomes (1,2); therefore, the need for additional education should be periodically assessed, particularly if treatment goals are not being met.

Training on the use of CGM often differs from that of pump and AID systems. In some settings, CGM training can be done by users themselves with company-provided training materials (for instance, people are able to use the over-the-counter [OTC] CGM brands on their own), although for underresourced, younger (8,9), and older (10) individuals, more repetition and time spent reviewing concepts is often indicated. Additionally, regular monitoring and review of the data obtained from CGM devices is needed to inform and optimize clinical care (8). Education and training should be available to any person who needs it but should not constitute a barrier to device use in a motivated individual.

Insulin pumps and AID systems, on the other hand, generally require training and education for safe use. There are studies where youth with type 1 diabetes have been able to self-initiate tubeless AID systems (11), but for most, training with a certified or trained diabetes educator and education specialist is necessary for full understanding and safe use of the technology.

Health care professionals should be aware of the availability of diabetes devices and their recommended uses and how to access the resources needed to use them. Given the rapidly changing landscape of technology, it is expected that software and/or hardware are being constantly updated; hence, periodic education is helpful to keep health care professionals informed and updated. For those interested in more advanced involvement with device use, there are defined competencies described as basic, fundamental, intermediate, and advanced that are specific to the role of each health care team member (12). The health care team's knowledge and competency are even more critical when people with diabetes are started on advanced diabetes technologies, such as AID systems. In such situations, training of the team is vital and should include a discussion about realistic expectations for the ability of the initiated system to achieve glucose goals, fundamental differences from the previous treatment tools, the system's features and limitations, and the best way to use the new system to maximize the benefits it can offer (13).

Use in Schools for Children and Adolescents

Technology use should be supported in school settings, and instructions for device use should be outlined in the student's diabetes medical management plan (DMMP). A backup treatment plan should be included in the DMMP for potential device failure. School nurses and designees should complete training to stay up to date on diabetes technologies prescribed for use in the school setting. Updated resources to support diabetes care at school, including training materials and a DMMP template, can be found online at (https://diabetes.org/advocacy/safe-at-school-state-laws/diabetes-medical-management-plan).

Use for Adults

For adults using devices in educational and employment settings, reasonable accommodations should be provided to allow time and space necessary for safe device use (14). This includes time to respond appropriately to high and low glucose levels and deal with any device notifications and issues.

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Initiation of Device Use

Diabetes technology should be initiated soon after the time of diagnosis and anytime thereafter, as indicated by the user's needs and wishes. The use of CGM and BGM devices should be started from the outset of the diagnosis of diabetes, particularly in people who require insulin for management (3,4,15,16). CGM use allows for close tracking of glucose levels with adjustment of insulin dosing and lifestyle modifications and removes the burden of frequent BGM. In addition, early CGM initiation after diagnosis of type 1 diabetes in children and adolescents has been shown to decrease A1C levels and is associated with high parental satisfaction and reliance on this technology for diabetes management (5,7). Training on alarm and alert settings when initiating CGM is crucial to avoid alarm fatigue or overload. Interruption of access to CGM is associated with a worsening of outcomes (6,17); therefore, it is important for individuals on CGM to have consistent access to devices. A recent systematic review with meta-analysis also supports the use of CGM in people with type 2 diabetes not using insulin, with observations of improved glycemic outcomes and individual experience, reduced health care resource utilization, and acceptable cost-effectiveness (18).

AID systems are the preferred insulin delivery system in individuals with type 1 diabetes and type 2 diabetes on multiple daily injections (MDI) and people with other forms of insulin-deficient diabetes. They can be considered for use in people on basal insulin who are not reaching their targets. Early initiation of AID therapy has been shown to be beneficial. In an open-label, multicenter, randomized, parallel clinical trial enrolling youth with newly diagnosed type 1 diabetes, initiation of an AID system within 21 days from diagnosis showed 10% higher time in range (TIR) (70–180 mg/dL [3.9–10.0 mmol/L]) and lower A1C at 12 months versus usual care (15). In addition, use of diabetes technology overall improves A1C and increases the number of people achieving an A1C <7% (19). There are no data that the presence or absence of C-peptide or islet autoantibodies is associated with responsiveness to continuous subcutaneous insulin infusion or AID therapy. People with type 2 diabetes who continue to produce insulin have been shown to benefit from AID therapy (20–23). Similarly, the duration of diabetes is not related to outcomes with the use of technology. Therefore, none of these measures should be used when determining suitability for continuous subcutaneous insulin infusion or AID therapy.

Standardized Reporting

Use of a standardized CGM tracing is helpful for people with diabetes and clinicians (24) and in multiple settings, from connected insulin pens (25) to CGM (26) to AID systems (27). Ideally, both people with diabetes and their health care teams can access and analyze the data, both between and at clinic visits to inform self-management and medication dose titration.

BLOOD GLUCOSE MONITORING

Recommendations

7.10 People with diabetes should be provided with blood glucose monitoring (BGM) devices as indicated by their circumstances, preferences, and treatment. People using CGM devices must also have access to BGM at all times. **A**

7.11 People who are taking insulin and using BGM should be encouraged to check their blood glucose levels when appropriate based on their insulin therapy. This may include checking when fasting, prior to meals and snacks, after meals, at bedtime, in the middle of the night, prior to, during, and after exercise, when hypoglycemia is suspected, after treating low blood glucose levels until achieving normoglycemia, when hyperglycemia is suspected, and prior to and while performing critical tasks such as driving. **B**

7.12 Health care professionals should be aware of the differences in accuracy among blood glucose meters. Only meters approved by the U.S. Food and Drug Administration (FDA) (or comparable regulatory agencies for other geographical locations) with proven accuracy should be used, with unexpired test strips purchased from a pharmacy or licensed distributor and properly stored. **E**

7.13 Although BGM in people on noninsulin therapies has not consistently shown clinically significant reductions in A1C levels, it may be helpful when modifying meal plans, physical activity plans, and/or medications (particularly medications that can cause hypoglycemia) in conjunction with a treatment adjustment program. **E**

7.14 Consider potential interference of medications and substances on glucose levels measured by blood glucose meters. **B**

Major clinical trials of insulin-treated people with diabetes have included BGM as part of multifactorial interventions to demonstrate the benefit of intensive glycemic management on diabetes complications (28). BGM is thus an integral component of effective therapy for individuals using insulin. In recent years, CGM has emerged as a method for the assessment of glucose levels that improves outcomes compared with BGM in many settings (discussed below). Glucose monitoring (BGM or CGM) allows people with diabetes to evaluate their individual responses to therapy and assess whether glycemic goals are being safely achieved. Integrating glucose testing results into diabetes management can be a useful tool for guiding medical nutrition therapy and physical activity, preventing hypoglycemia, or adjusting medications (particularly prandial insulin doses or correction bolus doses). The specific needs and goals of the person with diabetes should dictate BGM and CGM frequency and timing. As recommended by the device manufacturers and the U.S. Food and Drug Administration (FDA), people with diabetes using CGM must also have access to BGM for multiple reasons, including whenever there is suspicion that the CGM is inaccurate, while waiting for the CGM device to warm up, when there is a disruption in CGM transmission, for calibration (if needed) or if a warning message appears, when CGM supplies are delayed, and in any clinical setting where glucose levels are changing rapidly (>2 mg/dL/min), which could cause a discrepancy between CGM and blood glucose values.

Meter Standards

Glucose meters meeting FDA guidance for meter accuracy provide the most reliable data for diabetes management. There are several current standards for the accuracy of blood glucose meters,

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but the two most used are those of the International Organization for Standardization (ISO) (ISO 15197:2013) and the FDA (**Table 7.1**). In the U.S., currently marketed meters must meet the standard under which they were approved, which may not be the current standard. Moreover, the monitoring of current accuracy postmarketing is left to the manufacturer and not routinely checked by an independent source.

People with diabetes assume their glucose meter is accurate because it is FDA cleared, but that may not be the case. In prior analyses with glucose meters, 14 of 18 glucose meters met the minimum accuracy requirements (29). However, few studies have compared meters head-to-head.

Certain meter system characteristics, such as the use of lancing devices that are less painful (30) and the ability to reapply blood to a strip with an insufficient initial sample, or meters with integrated speech that can read aloud glucose levels for visually impaired individuals (31), may also be beneficial to people with diabetes (32) and may make BGM less burdensome to perform.

Use of Strips From Unlicensed Entities

People with diabetes should be advised against purchasing or reselling preowned or secondhand test strips, as these may give incorrect results. Only unopened and unexpired vials of glucose test strips should be used to ensure BGM accuracy.

Optimizing Blood Glucose Meter Use

Optimal use of BGM devices requires proper review and interpretation of data by both the person with diabetes and the health care professional to ensure that data are used in an effective and timely manner. In a large cohort analysis of over 24,000 adults with both type 1 and type 2 diabetes, more frequent use of BGM was associated with lower A1C levels, with greater BGM frequency associated with lower A1C in people with type 1 diabetes across the lifespan (33,34). In young adults with type 1 diabetes, there was also a correlation between greater BGM frequency and lower A1C levels (34). Among those who check their blood glucose at least once daily, many reported taking no action when results were high or low (35). Some meters now provide advice to the user in real time when monitoring glucose levels (36), whereas others can be used as a part of integrated health platforms (37). People with diabetes should be taught how to use BGM data to adjust food intake, physical activity, or pharmacologic therapy to achieve their treatment goals. The ongoing need for and frequency of BGM should be reevaluated at each routine visit to ensure its effective use (34,38).

People With Diabetes Treated With Intensive Insulin Therapies

Glucose monitoring is particularly important for people with diabetes treated with insulin therapy to detect and prevent hypoglycemia and hyperglycemia. Most individuals on intensive insulin therapies (MDI or insulin pump therapy) should be encouraged to assess glucose levels using BGM (and preferably CGM with BGM backup) prior to meals and snacks, at bedtime, occasionally postprandially, prior to, during, and after physical activity, when they suspect hypoglycemia or hyperglycemia, after treating hypoglycemia until they are normoglycemic, and prior to and while performing critical tasks such as driving. For many individuals using BGM alone, this requires doing finger sticks 6–10 times daily, although individual needs may vary. A database study of almost 27,000 children and adolescents with type 1 diabetes showed that, after adjusting for multiple confounders, increased daily frequency of BGM was significantly associated with lower A1C levels (−0.2% per additional check per day) and with fewer acute complications (39).

People With Diabetes Using Basal Insulin and/or Oral Agents and Noninsulin Injectables

The evidence is insufficient regarding when to prescribe BGM and how often monitoring is needed for insulin-treated people with diabetes who do not use intensive insulin therapy, such as those with type 2 diabetes taking basal insulin with or without oral agents and/or noninsulin injectables. However, for those taking basal insulin, assessing fasting glucose with BGM to inform dose adjustments to achieve blood glucose goals results in lower A1C levels (40).

In people with type 2 diabetes not taking insulin, studies have not consistently shown benefit, although the most benefit has been seen in individuals participating in structured clinical programs (41,42). Some individuals find BGM useful to provide insight into the impact of nutrition, physical activity, and medication management on glucose levels. Glucose monitoring may also be useful in assessing hypoglycemia, glucose levels during intercurrent illness, or discrepancies between measured A1C and glucose levels when there is concern an A1C

Table 7.1—Comparison of ISO 15197:2013 and FDA blood glucose meter accuracy standards

Setting	FDA*	ISO 15197:2013*
Hospital use	95% within 12% for BG ≥75 mg/dL 95% within 12 mg/dL for BG <75 mg/dL 98% within 15% for BG ≥75 mg/dL 98% within 15 mg/dL for BG <75 mg/dL	95% within 15% for BG ≥100 mg/dL 95% within 15 mg/dL for BG <100 mg/dL 99% in A or B region of consensus error grid‡
Home use	95% within 15% for all BG in the usable BG range† 99% within 20% for all BG in the usable BG range†	

BG, blood glucose; FDA, U.S. Food and Drug Administration; ISO, International Organization for Standardization. To convert mg/dL to mmol/L, see endmemo.com/medical/unitconvert/Glucose.php. *Data shown in the FDA column are from the FDA (191). Data shown in the ISO column are from the FDA (192). †The range of blood glucose values for which the meter has been proven accurate and will provide readings (other than low, high, or error). ‡Values outside of the "clinically acceptable" A and B regions are considered "outlier" readings and may be dangerous to use for therapeutic decisions (193).

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result may not be reliable in specific individuals (for more details, see section 2, "Diagnosis and Classification of Diabetes"). A key consideration is that performing BGM alone does not lower blood glucose levels. To be useful, the information must be integrated into clinical and self-management treatment plans. Further, as noted above, a recent systematic review with meta-analysis did support use of CGM in people with type 2 diabetes with benefits to glycemic management, individual experience, health care resource utilization, and cost-effectiveness (18,43).

Glucose Meter Inaccuracy

Although many meters function well under various circumstances, health care professionals and people with diabetes must be aware of factors that impair meter (and CGM) accuracy. A meter reading that seems discordant with the clinical picture needs to be retested or tested in a laboratory. Health care professionals in intensive care unit settings need to be particularly aware of the potential for incorrect meter readings during critical illness, and laboratory-based values should be used if there is any doubt. Some meters give error messages if meter readings are likely to be false (44).

Oxygen. Most currently available glucose monitors use an enzymatic reaction linked to an electrochemical reaction, either glucose oxidase or glucose dehydrogenase (45). Glucose oxidase monitors are sensitive to the oxygen available and should only be used with capillary blood in people with normal oxygen saturation. Higher oxygen tensions (i.e., arterial blood or oxygen therapy) may result in false low-glucose readings, and low oxygen tensions (i.e., high altitude, hypoxia, or venous blood readings) may lead to falsely elevated glucose readings. Glucose dehydrogenase–based monitors are generally not sensitive to oxygen.

Temperature. Because the reaction is sensitive to temperature, all monitors have an acceptable temperature range (45). Most will show an error if the temperature is unacceptable, but a few will provide a reading and a message indicating that the value may be incorrect.

Interfering Substances. There are several physiologic and pharmacologic factors that interfere with glucose readings measured with either personal blood glucose meters or professional blood glucose meters used in various inpatient settings (neonatal intensive care units, hospital wards, and intensive care units) (45). Meters vary in terms of their sensitivity to interfering substances, and this information should be available in the prescribing information for the meter. Some of the possible interfering substances are listed in **Table 7.2**.

CONTINUOUS GLUCOSE MONITORING DEVICES

See **Table 7.3** for definitions of types of currently available CGM devices.

Recommendations

7.15 Use of CGM is recommended at diabetes onset and anytime thereafter for children, adolescents, and adults with diabetes who are on insulin therapy, **A** on noninsulin therapies that can cause hypoglycemia, **C** and on any diabetes treatment where CGM helps in management. **C** The specific CGM device and method for use should be made based on the individual's circumstances, preferences, and needs. **E**

7.16 In people with diabetes on insulin therapy, CGM devices should be used as close to daily as possible for maximal benefit. **A** People with diabetes should have uninterrupted access to their supplies to minimize gaps in CGM. **A**

7.17 During pregnancy for individuals with type 1 diabetes, CGM can help achieve glycemic goals (e.g., time in range and time above range) **A** and A1C goal **B** and may be beneficial for other types of diabetes in pregnancy. **E** See section 15, "Management of Diabetes in Pregnancy," for more detail regarding use of technology in pregnancy.

7.18 In circumstances when consistent use of CGM is not feasible, consider periodic use of personal or professional CGM to adjust medication and/or lifestyle. **C**

7.19 Skin reactions, either due to irritation or allergy, should be assessed and addressed to aid in successful use of devices. **E**

7.20 People who wear CGM devices should be educated on potential interfering substances and other factors that may affect accuracy. **C**

CGM measures interstitial glucose (which correlates well with plasma glucose, although at times, it can lag if glucose levels are rising or falling rapidly). Although historically there were two basic types of personal CGM devices, real-time CGM (rtCGM) and intermittently scanned CGM (isCGM), now most available prescribed CGM devices have similar characteristics: no swiping required,

Table 7.2—Some of the more common interfering substances and/or conditions that affect blood glucose meters (for inpatient and outpatient use)

Substance or condition	Potential effects on glucose readings measured by BGMs*
Maltose†	Falsely higher blood glucose readings
Galactose	Falsely higher blood glucose readings
Xylose	Falsely higher blood glucose readings
N-Acetylcysteine	Falsely higher or lower blood glucose readings (depending on BGM design)
Acetaminophen	Falsely higher or lower blood glucose readings (depending on BGM design)
Dopamine	Falsely higher or lower blood glucose readings (depending on BGM design)
Pralidoxime (2-PAM)	Falsely higher or lower blood glucose readings (depending on BGM design)
Hydroxyurea	Falsely higher or lower blood glucose readings (depending on BGM design)
Vitamin C	Falsely higher or lower blood glucose readings (depending on BGM design)
Hematocrit (high)	Falsely lower blood glucose readings
Hematocrit (low)	Falsely higher blood glucose readings

*These are potential effects. There are blood glucose monitors (BGMs) that behave differently than listed in this table. Refer to product labeling for product-specific information. †Unmodified glucose dehydrogenase pyrroloquinoline quinone (GDH/PQQ) enzyme method only. Modern BGM designs do not incorporate unmodified GDH-PQQ enzyme.

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Table 7.3—Continuous glucose monitoring devices

Type of device	Brand*	Availability	Alarms
rtCGM	Libre 2 Plus and Libre 3 Plus	Prescription	Yes
	Dexcom G6 and G7	Prescription	Yes
	Eversense 365	Prescription	Yes
	Guardian 4	Prescription	Yes
	Simplera	Prescription	Yes
OTC-CGM	Dexcom Stelo	OTC	No
	Abbott Lingo	OTC	No
Professional CGM	Abbott FreeStyle Libre Pro	In office	No
	Dexcom G6 Pro	In office	No

CGM, continuous glucose monitoring; isCGM, intermittently scanned CGM; OTC, over the counter; rtCGM, real-time CGM. *Generic names not available.

continuous stream of data, and adjustable alarms and alerts. If an individual is still using a CGM that requires scanning, the limitations of those systems should be addressed. There are two categories of CGM devices generally available for personal use: those prescribed for diabetes management (rtCGM) and over-the-counter CGM (OTC-CGM), which lacks alarms and alerts and can be used by people not taking diabetes medications that increase the risk of hypoglycemia. Anyone can purchase the OTC-CGM devices, including those without diabetes or with prediabetes who wish to assess their glycemic responses to their lifestyles, including the effects of food choices and exercise. Finally, there are what has been called professional CGM devices, which are devices owned and applied by health care professionals that generate glucose data (blinded or not) over a short period of time.

Table 7.3 provides definitions for the types of CGM devices. For people with type 1 diabetes using CGM, frequency of sensor use is an important predictor of A1C lowering for all age-groups (46,47).

Few real-time systems require calibration by the user, and the need to recalibrate varies in frequency depending on the device. However, all CGM systems require use of BGM to confirm accuracy, especially in settings where there is perceived discordance between sensor reading and the individual's perceived sense of their clinical situation.

Most CGM systems are designated as integrated CGM (iCGM) and cleared as a class II medical device by the FDA for integration with other digitally connected devices. The exact CGM that integrates with any given device (e.g., automated insulin system or connected insulin pen) varies, and available options should be explored with individuals when choosing an integrated system.

Benefits of Continuous Glucose Monitoring

Data From Randomized Controlled Trials

Multiple randomized controlled trials (RCTs) have been performed using CGM devices, and the results have largely been positive in terms of reducing A1C levels and/or episodes of hypoglycemia if participants regularly wore the devices (46–53). The benefits of CGM have been shown regardless of age, sex, education or income levels, or baseline diabetes characteristics.

The initial studies were done primarily in adults, children, and adolescents with type 1 diabetes on insulin pump therapy and/or MDI (46,47,50,51,54). The primary outcome was met and showed benefit in adults of all ages (46,55,56), including older individuals (57–59). Data in children show that CGM use in young children with type 1 diabetes reduced hypoglycemia; in addition, behavioral support of parents of young children with diabetes using CGM showed the benefits of reducing hypoglycemia concerns and diabetes distress (46,51,60). Similarly, A1C level reduction was seen in adolescents and young adults with type 1 diabetes using rtCGM (50).

RCT data on CGM use in individuals with type 2 diabetes on MDI (61) or insulin (including basal insulin) plus noninsulin therapies (61–64) show consistent reductions in A1C levels and increases in TIR (70–180 mg/dL [3.9–10 mmol/L]) and time below range but not a reduction in rates of level 3 hypoglycemia (65). RCT data for CGM benefits in people with type 2 diabetes not using insulin are increasing and generally have shown greater benefits of CGM compared with BGM for A1C, TIR, time below range, and time above range (TAR) as well as greater user-reported satisfaction (66,67). In addition, CGM benefits were reported in a population of adults with type 2 diabetes (on noninsulin and insulin-including therapy) with reduction in A1C levels, increase in TIR, and reduction of time in hyperglycemia (>180 mg/dL [>10 mmol/L] and >250 mg/dL [>13.9 mmol/L]) (62). Although glycemic metrics are improved, patient-reported outcomes have shown less consistent satisfaction in this individual population.

Although short-term use of CGM in youth with type 2 diabetes did not impact short-term glucose changes or A1C improvement, users reported behavioral changes with increased blood glucose measurements, increased insulin administration, and overall improved diabetes management and quality of life (68,69). The improvements in type 2 diabetes have largely occurred without changes in insulin doses or other diabetes medications. CGM discontinuation in individuals with type 2 diabetes on basal insulin caused partial reversal of A1C reduction and TIR improvements, suggesting that continued CGM use achieves the greatest benefits (17).

Observational and Real-world Studies

CGM systems are widely available in many countries for people with diabetes, and this allows for the collection of large amounts of data across groups of people with diabetes.

Data for isCGM in adults with diabetes include results from observational studies, retrospective studies, and analyses of registry and population data (70,71). Glycemic benefits in people with type 1 and type 2 diabetes (on MDI, basal insulin, and noninsulin therapies) have been seen in prospective and retrospective studies (49,72).

Reductions in acute diabetes complications, such as diabetic ketoacidosis (DKA), episodes of severe hypoglycemia or diabetes-related coma, and hospitalizations for hypoglycemia and hyperglycemia, have been observed in adults with type 1 or type 2 diabetes (73–75), with persistent effects observed even after 2 years of CGM initiation (76). Similar reductions of acute diabetes events and all-cause inpatient hospitalizations were seen in a retrospective review of

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adults with type 2 diabetes treated with basal insulin or with noninsulin therapy 6 months after initiation of isCGM (77).

Retrospective data from rtCGM use in adults (78) with type 1 or type 2 diabetes treated with insulin showed that the use of rtCGM significantly lowered A1C levels and reduced rates of emergency department visits or hospitalizations for hypoglycemia but did not significantly lower overall rates of emergency department visits, hospitalizations, or hyperglycemia.

Recent data have emerged from a real-world observational analysis of rtCGM use in adults with type 2 diabetes not treated with insulin. In this study, rtCGM benefits were observed at 6 months and 12 months versus baseline, with reduction of mean glucose levels, reduction of glucose management indicator (GMI), increase in TIR, increase in time in tight target range (70–140 mg/dL [3.9–7.8 mmol/L]), and reduction in TAR >180 mg/dL (>10 mmol/L) and >250 mg/dL (>13.9 mmol/L) (79).

Real-time Continuous Glucose Monitoring Device Use in Pregnancy

CGM indication has expanded to include pregnancy for Dexcom G7, FreeStyle Libre 2, and FreeStyle Libre 3, which enhances care in this population (80,81). A discussion of CGM use in pregnancy can be found in section 15, "Management of Diabetes in Pregnancy."

Use of Professional Continuous Glucose Monitoring and Intermittent Use of Personal Continuous Glucose Monitoring

Professional CGM devices, which provide retrospective data, either blinded or unblinded, for analysis can be used to identify patterns of hypoglycemia and hyperglycemia (82–84). Professional CGM can be helpful to evaluate an individual's glucose levels when either rtCGM or isCGM is not available to the individual or they prefer a blinded analysis or a shorter experience with unblinded data. It can be particularly useful in individuals using agents that can cause hypoglycemia, as the data can be used to evaluate periods of hypoglycemia and make medication dose adjustments if needed. It can also be useful to evaluate periods of hyperglycemia.

Some data have shown the benefit of intermittent use of CGM (rtCGM or isCGM) in individuals with type 2 diabetes on noninsulin and/or basal insulin therapies (63,85). In these RCTs, people with type 2 diabetes not on intensive insulin therapy used CGM intermittently compared with those randomized to BGM. Both early (63) and late (63,86) improvements in A1C levels were found.

Furthermore, in a real-world study, the use of professional CGM in individuals with type 2 diabetes not on insulin at baseline and at 6 months of follow-up resulted in lower A1C at 6 months as well as a shift toward greater use of glucose-lowering medications with cardiometabolic benefits, such as sodium–glucose transporter 2 inhibitors and glucagon-like peptide 1 receptor agonists (87). Use of professional or intermittent CGM should always be coupled with analysis and interpretation for people with diabetes along with education, as needed, to adjust medication and change lifestyle behaviors (88–90).

Side Effects of Continuous Glucose Monitoring Devices

Contact dermatitis (both irritant and allergic) has been reported with all devices that attach to the skin (91–93). In some cases, this has been linked to the presence of isobornyl acrylate, a skin sensitizer that can cause an additional spreading allergic reaction (94). It is important to ask CGM users periodically about adhesive reactions, as tape formulations may change over time. Patch testing can sometimes identify the cause of contact dermatitis (95). Identifying and eliminating tape allergens is important to ensure the comfortable use of devices and promote self-care (96–99). The PANTHER Program offers resources in English and Spanish at www.panther-program.org/skin-solutions. In some instances, using an implanted sensor can help avoid skin reactions in those sensitive to tape (100,101).

Substances and Factors Affecting Continuous Glucose Monitoring Accuracy

Sensor interference due to several medications/substances is a known potential source of CGM sensor measurement errors (**Table 7.4**). While several of these substances have been reported in the various CGM brands' user manuals, additional interferences have been discovered after the market release of these products. Hydroxyurea, used for myeloproliferative disorders and hematologic conditions, is one of the most recently identified interfering substances that cause a temporary increase in sensor glucose values discrepant from actual glucose values (102,103). Similarly, substances such as mannitol and sorbitol, when administered intravenously or as a component of peritoneal dialysis solution, may increase blood mannitol or sorbitol concentrations and cause falsely elevated readings of sensor glucose (104). Therefore, it is crucial to routinely review the medications and supplements used by the person with diabetes to

Table 7.4—Continuous glucose monitoring device interfering substances

Medication	Systems affected	Effect
Acetaminophen		
>4 g/day	Dexcom G6, Dexcom G7	Higher sensor readings than actual glucose
Any dose	Medtronic Guardian 4	Higher sensor readings than actual glucose
Ascorbic acid (vitamin C), >500 mg/day	FreeStyle Libre 2, FreeStyle Libre 3	Higher sensor readings than actual glucose
Ascorbic acid (vitamin C), >1,000 mg/day	FreeStyle Libre 2 Plus, FreeStyle Libre 3 Plus	Higher sensor readings than actual glucose
Hydroxyurea	Dexcom G6, Dexcom G7, Medtronic Guardian 4	Higher sensor readings than actual glucose
Mannitol (intravenously or as peritoneal dialysis solution)	Senseonics Eversense365	Higher sensor readings than actual glucose
Sorbitol (intravenously or as peritoneal dialysis solution)	Senseonics Eversense365	Higher sensor readings than actual glucose

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identify possible interfering substances and advise them accordingly on the need to use additional BGM if sensor values are unreliable due to these substances.

INSULIN DELIVERY

Insulin Syringes and Pens

Recommendations

7.21 For people with insulin-requiring diabetes on multiple daily injections (MDI), insulin pens are preferred in most cases. Still, insulin syringes may be used for insulin delivery considering individual and caregiver preference, insulin type, availability in vials, dosing therapy, cost, and self-management capabilities. **C**

7.22 Insulin pens or insulin injection aids are recommended for people with dexterity issues or vision impairment or when decided by shared decision making to facilitate the accurate dosing and administration of insulin. **C**

7.23 Offer connected insulin pens for people with diabetes taking multiple daily insulin injections when appropriate. **B**

7.24 FDA-approved insulin dose calculators/decision support systems may be helpful for calculating insulin doses. **B**

Injecting insulin with a syringe or pen (105–111) is the insulin delivery method used by most people with diabetes (112), although inhaled insulin is also available. Others use insulin pumps or AID devices (see INSULIN PUMPS AND AUTOMATED INSULIN DELIVERY SYSTEMS, below). For people with diabetes who use insulin, insulin syringes and pens both can deliver insulin safely and effectively for the achievement of glycemic goals. Individual preferences, cost, insulin type, dosing therapy, and self-management capabilities should be considered when choosing among delivery systems. Trials with insulin pens generally show equivalence or small improvements in glycemic outcomes compared with using a vial and syringe. Many individuals with diabetes prefer using a pen because of its simplicity and convenience. It is important to note that while many insulin types are available for purchase as either pens or vials, others may be available in only one form or the other, and there may be significant cost differences between pens and vials (see **Table 9.4** for a list of insulin product costs with dosage forms). Insulin pens may allow people with vision impairment or dexterity issues to dose insulin accurately (113–115), and insulin injection aids are also available to help with these issues. (For a helpful list of injection aids, see diabetes.org/living-with-diabetes/treatment-care/diabetes-technology-guide). Inhaled technosphere insulin can be useful for people with diabetes, providing an alternative method of insulin delivery with very fast onset of action.

The most common syringe sizes are 1 mL, 0.5 mL, and 0.3 mL, allowing doses of up to 100 units, 50 units, and 30 units, respectively, of U-100 insulin. Some 0.3-mL syringes have half-unit markings, whereas other syringes have markings in 1- to 2-unit increments. In a few parts of the world, insulin syringes still have U-80 and U-40 markings for older insulin concentrations and veterinary insulin, and U-500 syringes are available for the use of U-500 insulin. Syringes are generally used once but may be reused by the same individual in resource-limited settings with appropriate storage and cleansing (115).

Insulin pens offer added convenience by combining the vial and syringe into a single device. Insulin pens, allowing push-button injections, come as disposable pens with prefilled cartridges or reusable insulin pens with replaceable insulin cartridges. Pens vary with respect to dosing increment and minimal dose, ranging from half-unit doses to 2-unit dose increments, with the latter available in U-200 insulin pens. U-500 pens come in 5-unit dose increments. Some reusable pens include a memory function, which can recall dose amounts and timing. Insulin pens, once started, can be kept in use for variable durations, based on the type of insulin, usually for 28 days, ranging from 14 to 56 days. Needle thickness (gauge) and length are other considerations. Needle gauges range from 22 to 34, with a higher gauge indicating a thinner needle. A thicker needle can give a dose of insulin more quickly, while a thinner needle may cause less pain. Needle length ranges from 4 to 12.7 mm, with some evidence suggesting that shorter needles (4–5 mm) lower the risk of intramuscular injection with erratic absorption and possibly the development of lipohypertrophy. When reused, needles may be duller and thus injections may be more painful. Proper insulin injection technique is a requisite for receiving the full dose of insulin with each injection. Concerns with technique and use of the proper technique are outlined in section 9, "Pharmacologic Approaches to Glycemic Treatment."

Connected insulin pens are insulin pens with the capacity to record and/or transmit insulin dose data. Insulin pen caps are also available and are placed on existing insulin pens and may assist with calculating insulin doses and by providing a memory function. Some connected insulin pens and pen caps can be programmed to calculate insulin doses, can be synced with select CGM systems, and can provide downloadable data reports. These pens and pen caps are useful to people with diabetes for real-time insulin dosing and allow clinicians to retrospectively review the insulin delivery times and, in some cases, doses and glucose data to make informed insulin dose adjustments (116). A quantitative study showed that people with diabetes preferred connected pens because of their ability to log insulin doses and glucose levels automatically (116). In a multicenter RCT in people with type 1 diabetes, the use of an insulin pen cap was associated with improved glycemic outcomes at 6 weeks in the insulin cap group, with an increase in TIR and decrease in GMI and TAR (117). A systematic review of connected insulin pens or pen caps showed improvement of glucose outcomes whether as A1C reduction, TIR increase, or hypoglycemia reduction (118). A recent real-world study with multinational data collected from 3,954 adults with diabetes using a connected pen and CGM validated the fact that treatment engagement with a connected insulin pen is positively associated with glycemic outcomes. On the other hand, missing as little as two basal doses or four bolus insulin doses over a 14-day period would be associated with a clinically relevant decrease in TIR of ≥5% (119).

Bolus calculators have been developed to aid dosing decisions (120–125). These systems are subject to FDA clearance to ensure safety and efficacy in terms of algorithms used and subsequent dosing recommendations. People interested in using these systems should be encouraged to use those that are FDA approved. Health care professional input and education can be helpful for setting the initial dosing calculations

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with ongoing follow-up for adjustments as needed.

Insulin Pumps and Automated Insulin Delivery Systems

Recommendations

7.25a AID systems are the preferred insulin delivery method over MDI, CSII, and sensor-augmented pumps in people with type 1 diabetes, **A** adults with type 2 diabetes, **A** children and adolescents with type 2 diabetes, **E** and those with other forms of insulin-deficient diabetes. **B**, **C**, **D**, **E** Choice of an AID system should be made based on the individual's circumstances, preferences, and needs. **E**

7.25b Consider AID systems for select people with type 2 diabetes treated with basal insulin not achieving individualized glycemic goals. **B** Choice of an AID system should be made basedon the individual's circumstances, preferences, and needs. **E**

7.26 Individuals with diabetes who have been using CSII and/or AID should have continued access across third-party payors. **E**

Stand-alone Insulin Pumps and AID Systems

Insulin pumps have been available in the U.S. for over 40 years. These devices deliver rapid-acting insulin throughout the day to manage glucose levels. Most insulin pumps use tubing to deliver insulin through a cannula, while a few attach directly to the skin without tubing (pods or patch pumps), and these systems have been approved for use in type 1 and type 2 diabetes. AID systems, which can adjust insulin delivery rates based on sensor glucose values, are preferred over nonautomated pumps and MDI in people with type 1 diabetes and have largely replaced the use of nonintegrated or standard insulin pumps. Recently, three AID systems were approved for use by people with type 2 diabetes.

Historically, studies that compared MDI with insulin pump therapy were relatively small and of short duration. However, a systematic review and meta-analysis concluded that pump therapy in people with type 1 diabetes has modest advantages for lowering A1C levels (−0.30% [95% CI −0.58 to −0.02]) and for reducing severe hypoglycemia rates in children, adolescents, and adults (126). Real-world data on insulin pump use in individuals with type 1 diabetes show benefits in A1C levels and hypoglycemia reductions as well as total daily insulin dose reduction (127).

Although AID systems have been shown to improve outcomes compared with MDI and insulin pump therapy alone, the choice of insulin delivery system is up to the individual along with support from their diabetes care team (126). Thus, the choice of insulin delivery method is based on the characteristics of the person with diabetes and which method is most likely to benefit them. Diabetes Wise (diabeteswise.org/), for individuals with diabetes, Diabetes Wise Pro (pro.diabeteswise.org/), for health care professionals, and the PANTHER Program (pantherprogram.org/device-comparison-chart) have helpful websites to assist health care professionals and people with diabetes in choosing diabetes devices based on their individual needs and preferences.

Adoption of AID and pump therapy in the U.S. shows geographical variations, which may be related to health care professional preference or center characteristics (128) and socioeconomic status, as pump therapy is more common in individuals of higher socioeconomic status, as reflected by private health insurance, family income, and education (128). Given the barriers to optimal diabetes care observed in disadvantaged groups (129), addressing the differences in access to insulin pumps and other diabetes technologies may lower health disparities (130).

AID systems preferentially, as well as insulin pump therapy alone or along with CGM, can be successfully started at the time of diagnosis (131) or at any point when they are needed in the course of an individual's diabetes. These devices can be safely used in adults (including older adults), adolescents, and children (15,16). Practical aspects of pump therapy initiation include assessment of readiness of the person with diabetes and their caregivers, if applicable. There is no consensus on which factors are most predictive of success with insulin pump or AID therapy (132). However, an understanding of the technical features of the system and how to troubleshoot if problems are encountered is necessary for safe use.

Complications and Challenges of Infusing Insulin

Complications of infused insulin can be caused by issues with infusion sets (dislodgement and occlusion), which put individuals at risk for ketosis and DKA and thus must be recognized and managed early (133). Other pump skin issues include lipohypertrophy or, less frequently, lipoatrophy (134) and pump site infection. Discontinuation of pump therapy is relatively uncommon today; the frequency has decreased over the past few decades, and its causes have changed (135). Current reasons for discontinuation are problems with cost or wearability, loss of insurance, dislike of the pump, suboptimal glycemic outcomes, or psychosocial considerations (e.g., anxiety or depression) (130). Common barriers to pump therapy adoption in children and adolescents are concerns regarding the physical interference of the device, discomfort with the idea of having a device on the body, therapeutic effectiveness, and financial burden (136–138).

Description of AID Systems

AID systems consist of mainly three components: an insulin pump, a CGM system, and an algorithm that determines insulin delivery. Based on the model and brand of currently FDA-approved and -cleared AID systems, the algorithm can be hosted in the pump body, in an insulin pod, or on a phone app. All AID systems on the market today integrate with one or more CGM systems and adjust insulin delivery either by modulating the preprogrammed basal rates or by replacing the basal rates with microboluses or microdoses of insulin every 5 min.

The modulation of insulin delivery is done by increasing, decreasing, or pausing insulin based on the CGM feedback, the predicted direction of the glucose levels, and the speed with which the glucose levels are changing. Different AID systems modulate insulin based on predicted glucose levels at various times, most commonly 30 min or 1 h. Currently available AID systems have either fixed glucose targets or adjustable glucose targets, ranging from 87 to 180 mg/dL (4.8 to 10 mmol/L), depending on the system. Glucose targets are generally set up for 24 h but can also be adjusted in some systems with up to eight segments per

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day. All current AID systems provide automated correction doses, whether embedded in the microdose adjustments every 5 min or by providing additional correction boluses with doses that are dependent on the various types of algorithms, with variable frequency and threshold glucose based on the type of control algorithm. Most AID systems can be used in manual mode, although this is generally not recommended, as the benefits of CGM modulation may be partially or totally lost. However, use of AID in manual mode may be necessary in some circumstances, therefore it is important to review and reassess manual-mode settings periodically. Current AID systems still require manual entry of carbohydrates for meal announcements or qualitative meal estimation announcements to calculate prandial doses.

Adjustments for physical activity are available in most AID systems currently on the market. These can be programmed in various time increments. In general, the glucose target is raised to prespecified levels based on AID systems, and these are often accompanied by more conservative insulin delivery to reduce the risk of hypoglycemia in the setting of increased insulin sensitivity other than physical activity, such as prolonged fasting or NPO status for procedures. Of note, some systems may still give autocorrection boluses if the glucose levels rise above a certain threshold even while the exercise/activity mode has been enabled. Details on the available AID systems and their features can be found at pantherprogram.org/device-type.

AID systems have largely replaced other methods of continuous subcutaneous insulin delivery due to the advantages they offer in insulin modulation to adjust insulin doses and minimize hypoglycemia and hyperglycemia.

Partial Closed-Loop Systems

Sensor-augmented pumps (SAPs) were the precursors of the currently used AID systems. They are discussed here more for historical perspective because they are no longer commonly used, having been replaced by more fully functional AID systems. SAPs consist of three components: an insulin pump, a CGM system, and an algorithm that automates insulin suspension when glucose is low or is predicted to go low within the next 30 min. Predictive low-glucose suspend systems have been shown to reduce time spent with glucose <70 mg/dL (<3.9 mmol/L) without rebound hyperglycemia during a 6-week randomized crossover trial in people with type 1 diabetes of various ages (139). Similar results were seen in additional studies in adults and children with reduction of hypoglycemia (140–142).

Data from Pivotal Trials

All currently FDA-approved and -cleared AID systems were tested for safety and efficacy in their pivotal trials in children, adolescents, and adults with type 1 diabetes (143–145) as well as in adults with type 2 diabetes (146,147). These studies were conducted by following a variety of methodologies, including observational studies as well as RCTs. Regardless of the study design, all AID system pivotal trials that examined individuals 2 years old or older, including older adults, have consistently demonstrated superiority to either standard insulin delivery or SAP and/or usual care (for the randomized trials), with consistent improvement in A1C, increase in TIR, especially overnight, as well as reduction of time spent in hypoglycemia (in the studies in people with type 1 diabetes). The greatest improvements were seen with AID when used in individuals with the highest baseline A1C or lowest TIR (148). These systems may also lower the risk of exercise-related hypoglycemia (149) and have been shown to have psychosocial benefits (150–155). A review of the literature on the health and economic value of AID systems found that AID systems are cost-effective (156). AID is the standard of care for people with type 1 diabetes who are capable of safely using the device and should be the preferred method of insulin delivery in these individuals. The decision to use AID systems should be made based on the preference of the person with diabetes (and/or caregivers) and the availability of resources to provide necessary training, access, and support for their use.

Data From Real-world Studies

Data from real-world studies on AID systems have become available and continue to increase rapidly. These studies include large numbers of users, at times even 30-fold higher than the number of people studied in AID pivotal trials (157). It is important to emphasize that for some AID systems all data are automatically collected to the database (158), whereas for other systems data are collected based on voluntary sharing to the database by AID users. A recent systematic review of AID real-world studies, with 20 studies representing 171,209 individuals, substantiated the results observed in the pivotal trials and have confirmed the clinical benefits of AID systems in people with type 1 diabetes. Newer systems have shown increased time spent in automation, and the real-world studies have retrospectively analyzed longer duration of system use compared with their respective pivotal trials, with most analyses occurring for more than 6 months and an average duration of 9 months (157). Benefits include improvement in A1C levels, TIR, and other glucometrics as well as psychosocial benefits (159–164).

Finally, real-world data showed that AID systems provide the same glycemic benefits to Medicare and Medicaid beneficiaries with type 1 and type 2 diabetes, emphasizing that access to this technology should be made available regardless of A1C levels and should be based on the individual's needs (165).

Automated Insulin Delivery Systems in Pregnancy

The use of AID systems in diabetes and pregnancy presents particular challenges, as the current FDA-approved AID systems (except for one that has been FDA approved but is not yet commercially available) have glucose goals that are not pregnancy specific and do not have algorithms designed to achieve pregnancy-specific glucose goals. Initiating or continuing AID systems during pregnancy needs to be assessed carefully. Selected individuals with type 1 diabetes should be evaluated as potential candidates for AID systems in the setting of expert guidance. The details on the use of AID systems in pregnancy are discussed in section 15, "Management of Diabetes in Pregnancy."

Insulin Pumps and Automated Insulin Delivery Systems in People With Type 2 and Other Types of Diabetes

Additional insulin delivery options in people with type 2 diabetes include disposable patch-like devices, which provide either a continuous subcutaneous insulin infusion of rapid-acting insulin (basal) with bolus insulin in 2-unit increments at the press of a button or bolus insulin only, delivered in 2-unit increments, used in

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conjunction with basal insulin injections (166–169). Use of a technology as a means of insulin delivery is an individual choice for people with diabetes and should be considered an option in those who are capable of safely using the device.

Open-Source Automated Insulin Dosing

Recommendation

7.27 Support and provide diabetes management advice to people with diabetes who choose to use an open-source AID system. **B**

Open-source automated insulin dosing (OS-AID) algorithms provide the precise code that governs their operation, so health care professionals and people with diabetes can have a more complete understanding of risks and benefits of their use (170). Any commercial entity could provide the source code for their interoperable automated glycemic controller, but most choose not to. OS-AID algorithms are largely designed, maintained, and curated by people with diabetes and their caregivers. Many people with diabetes use these algorithms with cleared CGM systems and insulin pump components. The information on how to set up and manage these systems is freely available online.

OS-AID is the preferred term when referring to any open-source system (commercial or otherwise). It is important to note that the term "DIY" is not reflective of any aspect of these community-driven systems. No individual person has written all the code for these algorithms, and a large percentage of users do not build the software themselves (171). There are two main available algorithms, the OpenAPS algorithm and the Loop algorithm, which have been implemented on a variety of platforms.

The OpenAPS heuristic algorithm (implemented on a system on a chip in OpenAPS, Android smartphones as AndroidAPS, and iPhone as iAPS/Trio) is supported by large real-world studies (172) and a multicenter RCT (173). The OpenAPS algorithm also supports unannounced meals. In a single-center study of adolescents with type 1 diabetes randomized to AndroidAPS with quantitative carbohydrate announcements, qualitative announcements, and no announcements, TIR was preserved across groups (174).

Loop, an open-source model predictive control algorithm, is implemented on iPhones as an app. Prospective real-world data from 558 adults and children with type 1 diabetes on this system (171) was used to support the FDA clearance of a variant called Tidepool Loop (175), now implemented commercially.

Both the Loop and OpenAPS algorithms offer direct management of algorithm aggressiveness through conventional pump settings. Therefore, it is advisable that health care professionals understand and offer support in tuning settings for these safe and effective technologies (170). This may include, for example, the adjustment of basal rates, insulin-to-carbohydrate ratios, or insulin sensitivity factors. As with any AID system, a backup insulin treatment plan is advisable.

Digital Health Technology

Recommendation

7.28 Consider combining technology (CGM, insulin pump, and/or diabetes apps) with online or virtual licensed coaching to improve glycemic outcomes in individuals with diabetes or prediabetes. **B**

Increasingly, people are turning to the internet for advice, coaching, connection, and health care. Diabetes, partly because it is both common and numeric, lends itself to the development of apps and online programs. Recommendations for developing and implementing a digital diabetes clinic have been published (176). The FDA approves and monitors clinically validated, digital, and usually online health technologies intended to treat a medical or psychological condition; these are known as digital therapeutics, or "digiceuticals" (fda.gov/medical-devices/digital-health-center-excellence/device-software-functions-including-mobile-medical-applications) (177). Other applications, such as those that assist in displaying or storing data, encourage a healthy lifestyle or provide limited clinical data support. Therefore, it is possible to find apps that have been fully reviewed and approved by the FDA and others designed and promoted by people with relatively little skill or knowledge in the clinical treatment of diabetes. There are insufficient data to provide recommendations for specific apps for diabetes management, education, and support in the absence of RCTs and validation of apps unless they are FDA cleared.

An area of particular importance is that of online privacy and security. Established cloud-based data aggregator programs, such as Tidepool, Glooko, and others, have been developed with appropriate data security features and are compliant with the U.S. Health Insurance Portability and Accountability Act of 1996. These programs can help monitor people with diabetes and provide access to their health care teams (178). Consumers should read the policy regarding data privacy and sharing before entering data into an application and learn how they can manage the way their data will be used (some programs offer the ability to share more or less information, such as being part of a registry or data repository or not).

Many online programs offer lifestyle counseling to achieve weight loss and increased physical activity (179). Many include a health coach and can create small groups of similar participants on social networks. Some programs aim to treat prediabetes and prevent progression to diabetes, often following the model of the Diabetes Prevention Program (180,181). Others assist in improving diabetes outcomes by remotely monitoring clinical data (for instance, wireless monitoring of glucose levels, weight, or blood pressure) and providing feedback and coaching (182–187). There are text messaging approaches that tie into a variety of different types of lifestyle and treatment programs, which vary in terms of their effectiveness (188,189). There are limited RCT data for many of these interventions, and long-term follow-up is lacking. However, in a real-world observational study of individuals with type 2 diabetes treated with basal insulin, noninsulin glucose-lowering medications, or no medications, the use of a digital health solution and rtCGM resulted in reductions of GMI and TAR >180 mg/dL (>10 mmol/L) and >250 mg/dL (>13.9 mmol/L) as well as an increase in TIR by 15% and participation in a least one engagement activity per week (190). Therefore, even with limited data, for an individual with diabetes, opting in to one of these programs can

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be helpful in providing support and, for many, is an attractive option.

Inpatient Care

Recommendations

7.29 In people with diabetes wearing personal CGM, the use of CGM should be continued when clinically appropriate during hospitalization, with confirmatory point-of-care glucose measurements for insulin dosing and hypoglycemia assessment and treatment under an institutional protocol. **B**

7.30 Continue use of insulin pump or AID in people with diabetes who are hospitalized when clinically appropriate. This is contingent upon availability of necessary supplies, resources, training, ongoing competency assessments, and implementation of institutional diabetes technology protocols. **C**

For details on technology use in the inpatient setting, see section 16, "Diabetes Care in the Hospital."

The Future

The pace of development in diabetes technology is extremely rapid. New approaches and tools are available each year. It is difficult for research to keep up with these advances, because newer versions of the devices and digital solutions are already on the market by the time a study is completed. The most important consideration in all these systems is the person with diabetes. Technology selection must be appropriate and safe for the individual. Simply having a device or application does not change outcomes unless the human being engages with it appropriately to create positive health benefits. This underscores the need for the health care team to assist people with diabetes in device and program selection and to support their use through ongoing education and training. Expectations must be tempered by reality—we do not yet have technology that completely eliminates the self-care tasks necessary for managing diabetes, but the tools described in this section can make it easier to manage if disseminated equitably.

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